From Critical Infrastructure to AI Factories: Building an AI Operations Copilot on Nebius Serverless AI

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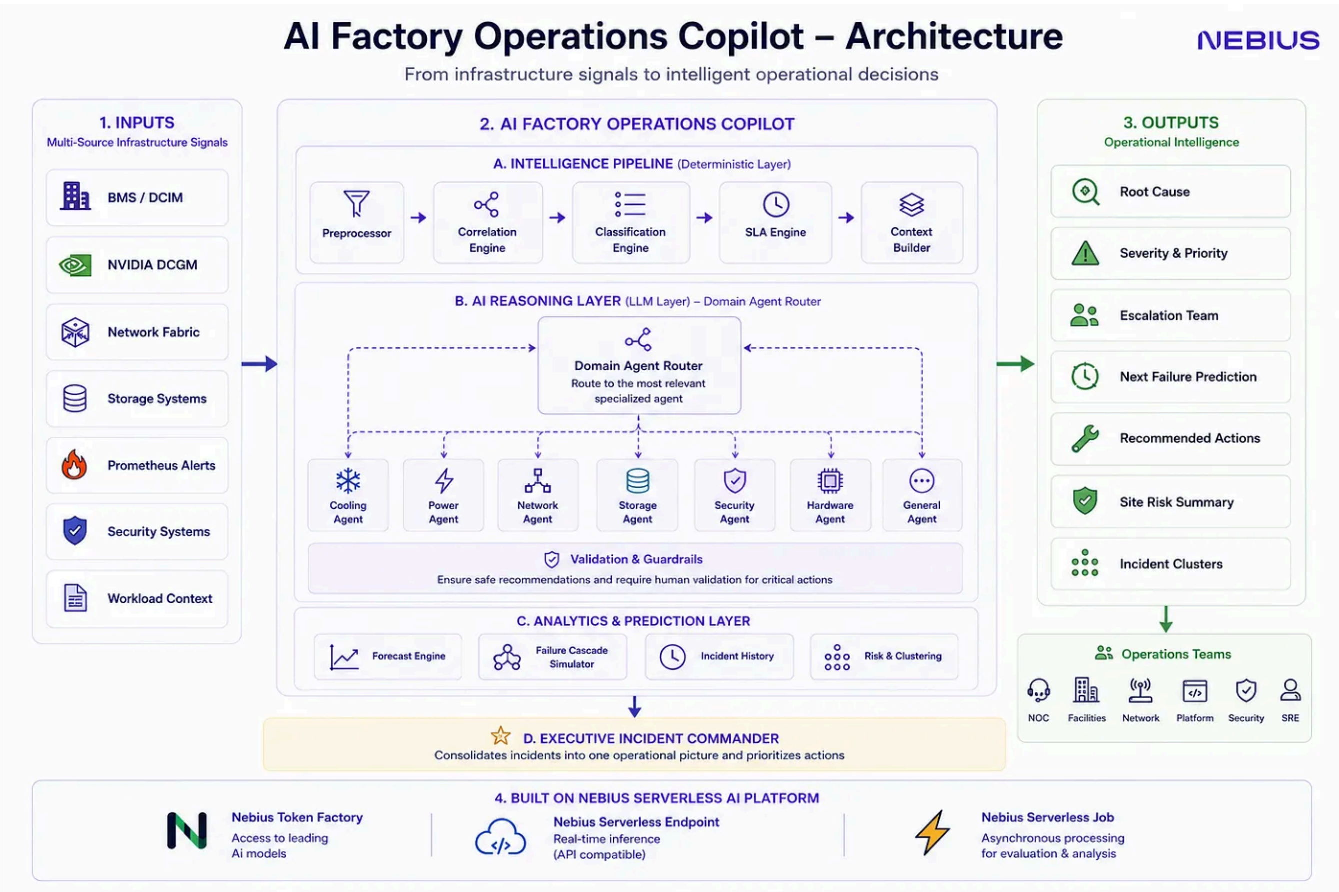
How experience in critical infrastructure, conversations with HPC professionals, and AI-assisted engineering inspired the design of an operational intelligence platform for AI factories.



ntroduction

The more I learned about AI infrastructure, the more I realized that incident response is becoming an information problem rather than simply a monitoring problem.

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Learning From Industry

One of the most valuable parts of this project happened outside the code editor.

During development I spoke with professionals working in large-scale infrastructure environments to better understand their operational challenges.

Those conversations completely changed my view of the project.

One recurring theme appeared again and again:

Operators already receive alerts.

What they need is context.

Another interesting insight came from discussing day-to-day work with engineers responsible for multiple vendors and infrastructure domains.

A single incident might require consulting documentation from server vendors, cooling vendors, networking vendors, storage vendors, or facility equipment manufacturers.

Finding the correct troubleshooting procedure often takes longer than understanding that something is wrong.

That inspired an important production vision for the project.

In a real deployment, each domain agent could evolve beyond prompt engineering into a specialist assistant backed by its own knowledge base, vendor documentation, operational runbooks, historical incidents, and retrieval systems.

Instead of answering from general model knowledge, each agent could reason using the organization’s own operational expertise and best practices.

Domain AI Agent

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Validation & Business Context

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Executive Incident Commander

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Operational Intelligence

Keeping the architecture simple was intentional.

I wanted someone unfamiliar with the codebase to understand the complete data flow within a minute.

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Safety Before Automation

One design decision became increasingly important as the project evolved.

Not every action should be automated.

Moving AI workloads between healthy GPU clusters is very different from opening a cooling valve or restarting a UPS.

The first may be safe automation.

The second could damage critical infrastructure if performed incorrectly.

Because of that, the Copilot intentionally separates:

- Safe workload automation
- Human-approved physical operations

The goal is not autonomous infrastructure.

Today's prototype demonstrates the overall architecture.

Each incident is routed to a specialized domain agent, but those agents currently share a common foundation model and differ primarily through domain-specific prompts.

In a production environment, I envision each domain agent becoming increasingly specialized.

Rather than relying only on prompts, every agent could have access to:

- Vendor documentation
- Operational runbooks
- Historical incident databases
- Organization-specific procedures
- Retrieval-Augmented Generation (RAG)
- Domain-specific fine-tuning where appropriate

Importantly, these agents would still operate **after** monitoring systems such as BMS, DCIM, Prometheus, and GPU telemetry have already collected and normalized infrastructure data.

The agents are not responsible for collecting sensor information.

They are responsible for interpreting operational context and assisting human decision-making.

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Wearing Multiple Hats

This project required much more than software development.

Throughout the challenge I found myself working simultaneously as:

- Product Manager
- Systems Architect

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Why Nebius?

Nebius Serverless AI was a natural fit for this project.

Operational incident analysis is bursty by nature.

Long periods of normal operation are interrupted by short bursts of intensive analysis.

Serverless infrastructure makes that model economically attractive while exposing a production-ready REST API with automatic scaling.

It also allows deterministic application logic and AI inference to remain cleanly separated.

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Lessons Learned

Building this project taught me that AI infrastructure is not only about GPUs.

It is about connecting physical infrastructure, operational processes, business priorities, and AI reasoning into one coherent system.

More importantly, I learned that the best AI systems are often not the ones making every decision.

They are the ones helping people make better decisions.

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Looking Ahead

The current implementation is intentionally a prototype.

Future work could include:

- Live Prometheus integration
- Real BMS and DCIM connectivity